Exam

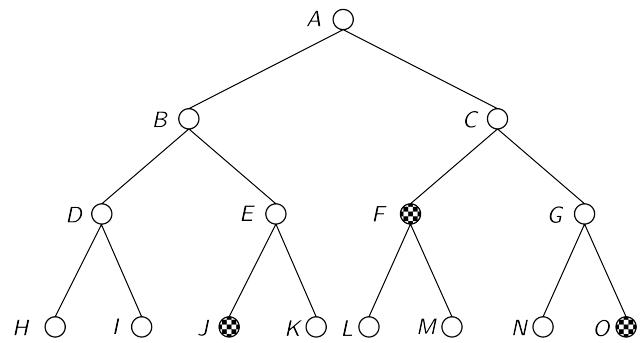
Instructions

- The exam will be graded out of 70 points, which will be combined with the 30 points from homework for your final score. The exam offers slightly more than 70 points, allowing you to choose which questions to answer.
- The total duration of the exam is 2 hours.
- The exam is divided into two sections: *direct application exercises*, which are straightforward questions based directly on the course material, and *problems*, which are more complex and open-ended.
- We do not expect you to fully solve the problems; partial answers can still earn points. We look for your ability to tackle unfamiliar challenges, propose coherent ideas, and discuss their limitations. However, answers without proper justification will not be awarded points.
- Please ensure your answers are clear and readable. Draft your ideas before writing them on the exam sheet. While minor corrections are acceptable, your answers must remain legible.
- The course content does not cover machine learning, except for Bayesian networks. Therefore, solutions involving other machine learning techniques (e.g., LLMs, deep learning algorithms, or reinforcement learning) will not be accepted.
- You have space to write your answers directly below each question. If you require additional space, use the blank sheets provided at the end and clearly reference the question number.
- Please avoid using red ink.
- Write your name on top of each page.

1 Direct application exercises (30 points)

1.1 Search in trees

Question 1 (3 points)**.** Consider the following search tree, where the goal states are indicated by a dashed node.



We consider an implementation in which an enumeration of the children of a node will be done from the left to the right (e.g. the enumeration of the children of A will yield B, C, in this order). Given that the algorithm stops when discovering a goal state, what is the sequence of nodes explored by:

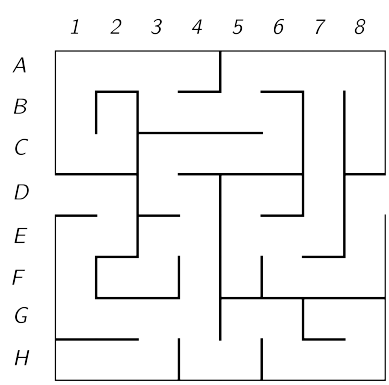
1. Depth First Search:

2. Breadth First Search:

3. Iterative Depth Lengthening:

1.2 Partial observability

Question 2 (3 points)**.** We consider an agent in a maze with partial observability: the agent can only see the walls in its current cell and knows its orientation (i.e. it knows exactly in which sides the walls are). The agent knows the map of the labyrinth, but ignores where it is located. It starts from a random cell, which is unknown to it.



Assuming that the agent is starting from cell $H5$, what is its current belief state? Write the transition for playing the Up action.

1.3 Game theory

In the following questions, Player 1 plays the rows and Player 2 the columns.

Question 3 (2 points)**.** What are the pure Nash equilibria of the following bi-matrix game?

$$G = \begin{pmatrix} 0,2 & 8,1 & 4,3 & 2,3 \\ 5,1 & 2,8 & 1,5 & 7,4 \\ 4,3 & 3,4 & 4,1 & 3,5 \end{pmatrix}$$

Question 4 (3 points)**.** What are the mixed Nash equilibria of the following bi-matrix game?

$$G = \begin{pmatrix} 1,-1 & 2,1 \\ 3,2 & 1,0 \end{pmatrix}$$

1.4 Logic

Question 5 (2 points)**.** If a, b and c are propositional variables, is the following sentence a tautology? Unsatisfiable? Justify your answer.

$$((a \to b) \wedge c) \leftrightarrow ((\neg c \to b) \wedge \neg a)$$

Question 6 (4 points)**.** If ϕ and ψ are two propositional formulae, prove that

$$\phi \to \psi \quad \equiv \quad \neg\phi \vee \psi$$

Question 7 (3 points)**.** Given the predicates *parent(x, y)* (testing that x is a parent of y) and $siblings(x, y)$ (testing that x and y are siblings), write in first-order logic an expression for the predicate $cousin(x, y)$, testing that x and y are cousins.

Question 8 (2 points)**.** In description logics, we consider the following interpretation $\mathcal{I}$:

- $\mathcal{D}^{\mathcal{I}} = \{e_1, e_2, e_3, e_4, e_6\}$
- Concepts: $A^{\mathcal{I}} = \{e_1, e_2, e_6\}$, $B^{\mathcal{I}} = \{e_2, e_4\}$, $C^{\mathcal{I}} = \{e_1, e_3, e_5\}$, and $D^{\mathcal{I}} = \{e_2, e_4, e_6\}$
- Roles: $r^{\mathcal{I}} = \{(e_1, e_3), (e_2, e_4), (e_3, e_6)\}$, $s^{\mathcal{I}} = \{(e_5, e_1), (e_3, e_4), (e_2, e_2)\}$, and $t^{\mathcal{I}} = \{(e_2, e_3), (e_3, e_4), (e_5, e_5)\}$

For each of the following concepts $\mathcal{C}$, what is the corresponding set $\mathcal{C}^{\mathcal{I}}$?

1. $A \sqcap C$

2. $\exists r.D$
3. $\forall s.B$
4. $\exists t.\forall r.(A \sqcup B)$

Question 9 (1 point)**.** If we are building a modal logic in which $\Box\phi$ interprets as "The agent has a proof of ϕ". How would one interpret $\Diamond\phi$?

1.5 Knowledge engineering

Question 10 (1 point)**.** Explain briefly the difference between an ABox and a TBox.

Question 11 (3 points)**.** What are three main limitations of RDFS?

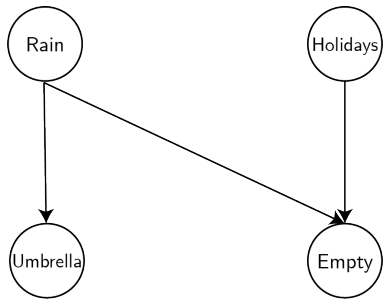
1.6 Probabilistic reasoning

Question 12 (3 points)**.** When walking in the street, Alice has to decide whether to take an umbrella. This decision is motivated mostly by the fact that it would be raining: Alice takes her umbrella with probability 0.9 if it is raining, and with probability 0.2 if it is not. However, our agent cannot directly observe whether it is raining. It can only observe the number of people outside. This is an imperfect source of information though, because an empty street can be caused not only by the weather, but also being or not holiday time

(which is the case with probability 0.15):

$$P(Empty \mid Rain, Holidays) = 0.7$$
$$P(Empty \mid \neg Rain, Holidays) = 0.5$$
$$P(Empty \mid Rain, \neg Holidays) = 0.6$$
$$P(Empty \mid \neg Rain, \neg Holidays) = 0.25$$

This is described using the following Bayesian network:



Compute the probability $P(Umbrella|Empty)$ that Alice takes her umbrella, given that we observe an empty street. You do not need to do the numerical computation, but you must give an expression that depends on the quantities introduced in the question, as well as a non-ambiguous way to compute the probability if you had access to a calculator.

1.7 Natural Language Processing

Question 13 (2 points)**.** Consider an alphabet $\mathcal{A} = \{a_1, \ldots, a_n\}$. We call a *palindrome* any string on $\mathcal{A}$ that is similar when read from the left to the right and from the right to the left. For instance "bob" is a palindrome, but "obob" is not. Propose a context-free-grammar for palindromes. You can consider that your grammar already has some lexicon for letters, of the form $L \to a_1 \mid \ldots \mid a_n$.

2 Problems (46 points)

2.1 The hawk-dove game (8 points)

The hawk-dove game is a 2-player game modeling a situation where two agents are fighting for some food. They can adopt two (pure) strategies: either they fight to get as much as possible (**H**awk) or they are more passive (**D**ove).

When two doves meet, they share the food equally. When two hawks meet, they fight for the food, and consequently waste a bit of it. When a hawk meets a dove, the hawk fights to get more of the food, but there is no waste.

We denote by α the total proportion of food left after the fight of two hawks (if $\alpha = 1$, there is no waste and they split the food equally; if $\alpha = 0$, all the food is lost in the fight). We denote by β the amout of food gained by the hawk when fighting against a dove (if $\beta = 1$, the hawk takes all the food).

The gain of each player is the total proportion of food they get. We can then represent the hawk-dove game as the following normal form game:

		P2	
		H	D
P1	H	$\alpha/2, \alpha/2$	$\beta, 1-\beta$
	D	$1-\beta, \beta$	$1/2, 1/2$

Question 14. Compute the Nash equilibria of the hawk-dove game depending on the values of α and β.

2.2 Backward chaining (10 points)

We consider the domain of propositional logic. Backward chaining is a proving algorithm used to verify whether a proposition ϕ is entailed by a knowledge base $\mathcal{K}$. The idea of the algorithm is to start from the target proposition ϕ and to retrieve all the premises that could make the proposition true following some inference rule.

For instance, if $\phi = a \vee b$, a premise could be a, with the inference rule:

$$\frac{a}{a \vee b}$$

Question 15 (2 points)**.** Write down the modus ponens inference rule, as well, as 2 additional inference rules of your choice.

Question 16 (3 points)**.** Show how backward chaining can be used to prove $a \vee (b \rightarrow c)$ based on the knowledge base $\mathcal{K} = \{d \wedge e, d \rightarrow \neg b\}$.

Question 17 (5 points)**.** Express backward chaining as a search problem. Describe the full specification of the search problem, explain whether the search is done on a tree, an And-Or tree or a graph, whether it is fully observable or partially observable etc. Which algorithm would you use to do the search?

2.3 Describing Pokemon games (10 points)

Following is an adaptation of the Wikipedia page on the Pokemon games:

Pokemon is a series of video games developed by Game Freak and published under the Pokemon media franchise. It was created by Satoshi Tajiri. The first games, Pocket Monsters Red and Green, were released in Japan, amd then outside of Japan as Pokemon Red Version and Blue Version. The most recently released core series games are Pokemon Scarlet and Violet. In the Pokemon games, the main player possesses several fictional creatures called "Pokemon". The player has to catch some wild Pokemon, train their own Pokemon and use them to battle other trainers' Pokemon.

Question 18. Propose a knowledge description of the domain as an OWL ontology. You can either represent it as a graph or with the RDF syntax.

Remark: Out of the 10 points, 4 are given for the correctness of the representation, 4 for using the right syntax and 3 for using specific properties of RDFS and OWL.

2.4 Korean alphabet (8 points)

Modern Korean alphabet (called *hangul*) uses 24 basic letters: 14 consonants and 10 vowels. The letters are not written one after the other like we do in the Latin alphabet, but grouped by syllabic blocks, in which the letters are arranged in two dimensions.

A base syllable is made up of one consonant and one vowel. If the symbol of the vowel is horizontal, the vowel is written below the consonant. Otherwise it is written to the right of the consonant. A second consonant can be added at the end of the syllable, and is written below the block made up of the first consonant and the vowel.[1]

Question 19. Propose a description of syllables in the Korean alphabet using the description language of your choice. Your description must assess the relative positions of the letters in the syllable. You can introduce the assertions of your choice, reflecting the nature of each individual letter (consonant, vowel, horizontal, vertical etc).

[1] The rules are actually slightly more complicated, but the essence remains well described in this simplification.

2.5 Wordle (10 points)

Wordle is a word game, the goal of which is to guess a 5-letter word in at most 6 attempts. After each attempt, the player is given a feedback for each letter, indicating whether:

- the letter is correct (displayed in green in the interface);
- the letter is incorrect but is located in another position in the word (displayed in yellow in the interface); or
- the letter is incorrect and not present in the word (displayed in gray in the interface)

Question 20 (10 points)**.** Assuming that you have access to a dataset $\mathcal{W}$ of candidate words, propose a formalization of the Wordle problem and a method to solve it.